

CURRICULUM VITAE

Aaron Kuang

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Citizenship: USA

EDUCATION

University of California, Irvine (UCI)

B.S. in Computer Science

Campuswide Honors Collegium, ICS Honors Program.

Honors Thesis: Help-seeking Behaviors in Algorithms Courses

Mar 2026

GPA: 4.00/4.00

RESEARCH INTERESTS

Computing Education

Interested in discovering and alleviating barriers to success, individual and societal, in computing. Particularly curious about how psychological factors — such as sense of belonging, self-efficacy, and impostor syndrome — manifest in computing students. Emphasis on incorporating student sentiments, boosting engagement, fostering curiosity, and broadening participation in computing, especially in non-traditional paths and subfields.

Human–AI Interaction

Interested in understanding how people — especially students, teachers, creatives, and researchers — interact with and perceive generative AI. Emphasis on issues related to ethics, cognitive offloading, and effects of integration. Investigating how generative AI has transformed learning environments and changed learning outcomes.

Algorithm Design

Curious about designing algorithms for computing paradigms beyond classical frameworks, such as parallel and quantum algorithms, with an emphasis on producing intuitive explanations and conveying them in a simple, instructive, and engaging manner.

RESEARCH EXPERIENCE

Help-seeking Behaviors in Algorithms Courses

Sep 2025 – Present

Advisor: Prof. Michael Shindler

Co-PI/Mentor: Prof. Stephen Tsung-Han Sher

- Helped conduct a qualitative research study that found that students enrolled in algorithms courses at one public R1 university weighed trade-offs between convenience, quality, and perceived effect on grades when seeking help — paper in preparation.
- Coordinated and scheduled 22 interviews between members of the research group and study participants by collecting interviewer and interviewee availability; personally conducted five of these semi-structured interviews.
- Oversaw the anonymization and cleaning of student transcripts, then assisted with qualitative coding, phenomenographic analysis, and paper writing.
- Attended two weekly meetings to review literature, debrief on current progress, identify cross-cutting trends seen in the transcripts, and discuss paper submission.

- Wrote an independent literary analysis research paper examining how the technological dystopia of *Fahrenheit 451* reduces its citizens to mere cogs in a machine.
- Discussed how anti-intellectualism and over-reliance on technology has eroded the characters' need to think critically, rendering them blind to the horrors of the world.
- Reflected on how modern feats of technology — social media, virtual reality, and artificial intelligence — can erase our individuality and humanity if misused.
- Stressed a need to inform technological innovation with knowledge from the humanities to ensure that technology enhances, rather than hinders, the human experience.

TEACHING EXPERIENCE

Learning Assistant (University of California, Irvine):

Design and Analysis of Algorithms

Spring 2025, Fall 2025

Data Structure Implementation and Analysis

Fall 2024

- Facilitated active learning in lectures and discussions by supporting students during in-class activities designed to reinforce concepts in data structures and algorithms.
- Supported exam review sessions meant to help students with recursion, graphs, and proofs. Presented slideshows and provided study tips.
- Attended weekly meetings with course staff to discuss pedagogical strategies, potential weak points in students' understanding, and course logistics.
- Helped administer exams in the algorithms course — proctored and scanned exams.

AWARDS, HONORS, AND SCHOLARSHIPS

Outstanding Contribution to Research by an ICS Undergraduate Jun 2026

- Nominated by a UCI faculty member for contributions to research.

UCI PBK Fall Book Award

Nov 2023

- Awarded to UCI sophomores who demonstrate academic excellence.

Victoria Alegria Tracy Endowed Scholarship

Sep 2023

- \$3,000 awarded to four UCI undergraduates annually.

UROP Humanities Core Research Paper Award Nomination

Jun 2023

- Nominated for writing a “superior research paper” on *Fahrenheit 451*.

AFFILIATIONS

Phi Beta Kappa

Since Jun 2026

- Nominated by UCI's Mu chapter as part of the top 5% of graduating seniors.

UCI ICS Honors Program (ICSHP)

Sep 2025 – Mar 2026

- Provides an opportunity for undergraduate ICS majors to be involved in research.

UCI Campuswide Honors Collegium (CHC)

Sep 2022 – Mar 2026

- Community of learners dedicated to scholastic excellence and personal growth.

OTHER ACTIVITIES/INTERESTS

Other Activities: Assisting with course schedules, Giving research presentations, Up-lifting and mentoring underclassmen, Auditing weekly algorithms research seminars.

Other Interests: Dystopian fiction, Creative writing, Literary analysis/critique, Astronomy, Pedagogical theory, Algorithm visualization, Sudoku, Cultivating desire paths.